⚙ Configuration

Upload Energy CSV File

Drag and drop file here

Limit 200MB per file • CSV

Browse files

building_energy_80... ×

55.6MB

Cost per Energy Unit ($)

0.12 − +

Select Anomaly Detection Model

Isolation Forest

Run Analysis

⚡ Energy Anomaly Detection System

AI-Powered Commercial Energy Intelligence Platform

Developed by Sagar Karosiya

Preprocessing Results

Missing Values

0

Min Scaled Value

0.0

Timestamp Sorted

True

Max Scaled Value

1.0

⚙ Feature Engineering Results

Total Features After Engineering: 52

Total Features After Engineering: Index(['timestamp', 'electricity', 'chilled_water', 'steam', 'hot_water', 'gas', 'water', 'temperature', 'humidity', 'hour', 'day', 'month', 'dayofweek', 'week', 'quarter', 'is_weekend', 'hour_sin', 'hour_cos', 'month_sin', 'month_cos', 'roll_mean_6', 'roll_std_6', 'roll_min_6', 'roll_max_6', 'roll_mean_12', 'roll_std_12', 'roll_min_12', 'roll_max_12', 'roll_mean_24', 'roll_std_24', 'roll_min_24', 'roll_max_24', 'roll_mean_48', 'roll_std_48', 'roll_min_48', 'roll_max_48', 'roll_mean_168', 'roll_std_168', 'roll_min_168', 'roll_max_168', 'lag_1', 'lag_2', 'lag_3', 'lag_6', 'lag_12', 'lag_24', 'lag_48', 'lag_72', 'lag_168', 'lag_336', 'zscore_24', 'zscore_168'], dtype='object')

✓ Rolling deviation metrics added

✓ Temporal seasonality features added

✓ Lag & momentum features added

Model Results

Anomaly Rate (%)

5.0

Total Anomalies

39984

Model Used

Isolation Forest

Anomaly Rate (%)

5.0

✓ Anomaly labels generated

✓ Anomaly predictions generated

✓ Anomaly scores computed

✓ Model Competitable with Isolation Forest

✓ Model Competitable with Local Outlier Factor (LOF)

✓ Model Competitable with One-Class SVM

✓ Model Competitable with Robust Covariance

Data Overview

	Column Name	Data Type
0	timestamp	datetime64[ns]
1	electricity	float64
2	chilled_water	float64
3	steam	float64
4	hot_water	float64
5	gas	float64
6	water	float64
7	temperature	float64
8	humidity	float64
9	hour	int32

Evaluation Results

Total Samples

799664

Anomaly Rate (%)

5.0

Total Anomalies

39984

✓ Anomaly distribution statistics calculated

✓ Top anomalous samples identified

Top 10 Most Anomalous Samples

	timestamp	electricity	chilled_water	steam	hot_water	gas	water	temperature	humidity	hour	day	month	dayofweek	week	q
183207	2036-11-24 15:00:00	1	0.484	0.7101	0.7041	0.6417	0.2567	0.644	0.7684	15	24	11	0	48	
565611	2080-07-10 03:00:00	0.2009	0.8675	0.6182	0.8979	0.4509	0.7097	0.375	0.5857	3	10	7	2	28	
692210	2094-12-19 02:00:00	0.0849	0.8592	0.5894	0.8322	0.386	0.554	0.5938	0.8117	2	19	12	6	50	
228719	2042-02-02 23:00:00	0.0942	0.6557	0.7661	0.2669	1	0.9998	0.5143	0.2385	23	2	2	6	5	
316741	2052-02-18 13:00:00	0.0121	0.3907	0.708	0.7339	0.568	0.6686	0.9224	0.3977	13	18	2	6	7	
693214	2095-01-29 22:00:00	0.1477	0.9226	0.1299	0.6978	0.5492	0.7407	0.4953	0.7172	22	29	1	5	4	
473991	2070-01-26 15:00:00	0.4578	0.5823	0.7865	1	0.3905	0.5935	0.5911	0.8834	15	26	1	6	4	
746818	2101-03-13 10:00:00	0.4185	0.4964	0.5484	1	0.6207	0.5989	0.562	0.4813	10	13	3	6	10	
667109	2092-02-07 05:00:00	0.0958	0.6301	0.2214	0.3301	0.347	0.6523	0.4786	0.7004	5	7	2	3	6	
96050	2026-12-16 02:00:00	0.316	1	0.6709	0.8894	0.7121	0.719	0.4894	0.5377	2	16	12	2	51	

Business Insight & Impact

Estimated Cost Impact ($)

2699.9

Anomaly Rate (%)

5.0

Peak Hour Analysis

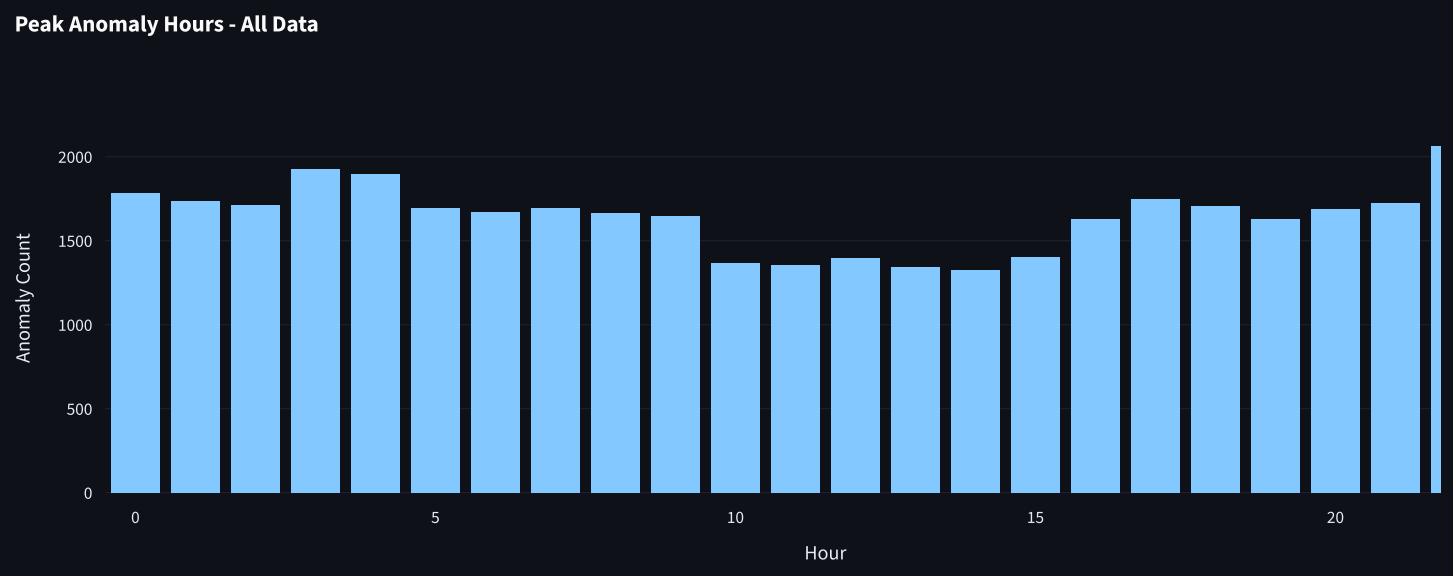
Select Month

All

Select Specific Date

All

Peak Anomaly Hours (All Data)



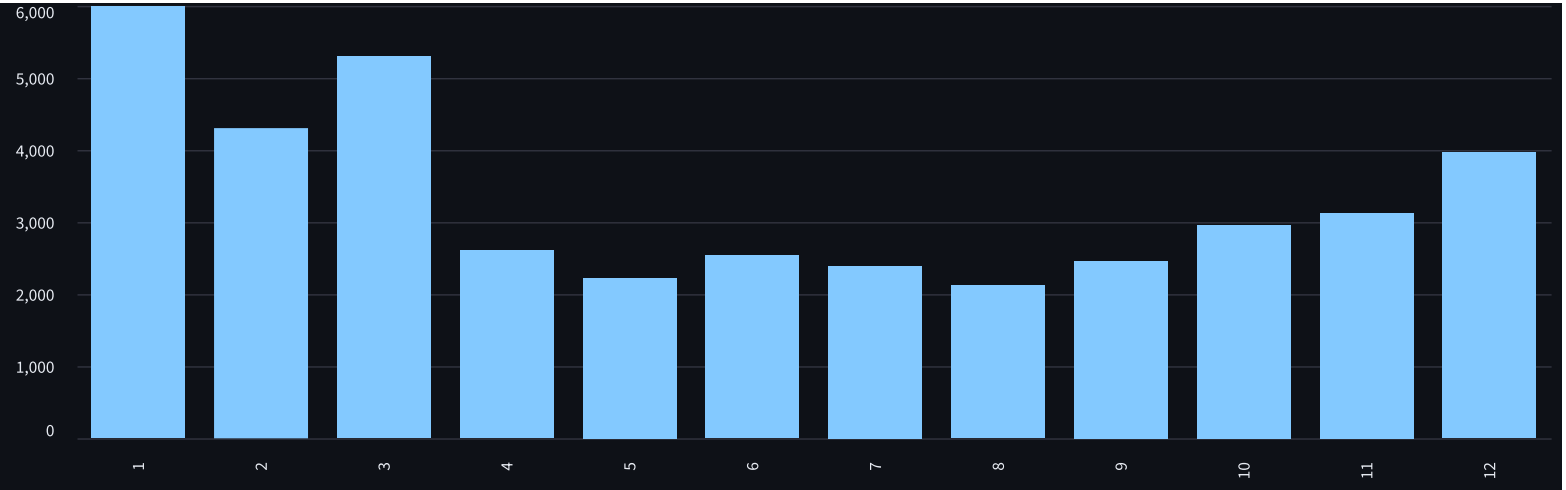
Peak Hour Summary

Analysis Scope: **All Data**

Peak Anomaly Hour(s): **23:00**

Maximum Anomalies at that hour: **2201**

Seasonal Pattern (All Data)



Recommendations

System operating within acceptable anomaly range.

Continue monitoring for early detection of faults.

Moderate cost impact. Optimize peak-hour operations.

Peak anomalies frequently occur around 23:00. Inspect load conditions during this time window.

Download Executive Report

Download PDF Report

© 2026 Sagar Karosiya | Energy AI Dashboard
All Rights Reserved